Stabilitest: Robustness and Fragility Analysis of Statistical Test Conclusions in R

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Abstract A p-value quantifies how surprising the observed data would be under the null hypothesis, but not how easily the resulting decision could be overturned. The stabilitest package combines three sensitivity views of one pre-specified analysis: jackknife leave-one-out influence, greedy worst-case observation removal, and the bootstrap same-decision rate. Component metrics and a composite 0-100 score are returned for two-sample location and proportion tests, linear model and ANCOVA terms, GLM and Cox terms, and TOST equivalence and non-inferiority endpoints. Categorical interpretation is fail-closed: a label is emitted only where an independent pre-registered study validates thresholds for that exact configuration. A Fisher exact-test study validated a two-band mapping. A prospective Welch study completed 4,500 training analyses but found no feasible mapping, so held-out data remained unopened and Welch labels are suppressed. The article also reports negative ANCOVA calibration studies. These outcomes illustrate why numeric sensitivity diagnostics and categorical assurance claims require different evidentiary standards.

1 Introduction

Statistical reporting in clinical trials typically compresses the evidence into a binary decision at $p < \alpha$. That compression discards something a reader needs. Consider two trials that both report $p = 0.048$. In the first, the conclusion survives essentially any plausible perturbation of the data. In the second, it rests entirely on one extreme responder: delete that single patient and significance evaporates. Standard reporting treats these as identical findings.

This is not a hypothetical concern: sensitivity of conclusions to individual patients is a recurring theme in the trial-reporting literature (Senn, 2007; Pocock et al., 2015). The ICH E9 guideline and its E9(R1) addendum (International Council for Harmonisation, 1998, 2019) make it explicit, asking that inferences be robust to data limitations and to deviations from the assumptions of the main estimator, assessed through pre-specified sensitivity analyses. What the guidance does not supply is quantitative machinery. Three distinct questions jointly characterise the stability of a test conclusion, and each needs a different tool:

1. *Which individual observations drive the result?* (influence)
2. *What is the smallest set of observations whose removal would overturn the conclusion?* (fragility)
3. *How often do bootstrap resamples preserve the observed decision?* (same-decision rate)

stabilitest (Ardelean, 2026) answers all three for one pre-specified primary analysis, returning component metrics plus a composite score. Its contributions are:

- **An integrated framework.** Jackknife influence, adversarial worst-case removal, and the bootstrap same-decision rate are computed together for a common catalogue of trial-relevant analyses — two-sample location and proportion tests, linear model and ANCOVA terms, GLM and Cox terms, and TOST equivalence and non-inferiority — through a small, uniform API.
- **A fail-closed calibration registry.** Categorical labels (*Robust* / *Fragile*) are treated as claims that require evidence. The package emits a label only where an independent, pre-registered calibration study has validated thresholds for that exact method, endpoint, conclusion type, weight vector, and design profile. Everywhere else the label is `NA` while numeric output is retained. Thresholds are never transferred between method families by analogy.
- **A calibrated unit, earned.** An independent study for Fisher's exact test yielded a validated two-band mapping, reported here with its held-out confirmation bounds and a descriptive comparison with a p-value cutoff.
- **Named negative results.** Two prospective Welch studies and three ANCOVA studies closed without usable thresholds under their frozen designs. These empirical failures are reported rather than converted into weaker claims.

Deletion sensitivity is complementary to the estimand and sensitivity-analysis strategy required by ICH E9(R1). It identifies dependence on observed rows; it does not by itself address missing-data assumptions, intercurrent events, or alternative estimands.

Table 1: R packages addressing related sensitivity questions. Detailed endpoint and method comparisons are supplied in related-packages.md. Entries were checked against CRAN or repository documentation on 2026-08-10; fragilityindex is excluded because it has been archived on CRAN since 2020-07-08.

Package	Threat	Scope
stabilitest	Observed-row deletion	Tests and model terms
fragility	Outcome-status flips	Binary outcomes
sensemakr	Omitted confounding	Linear regression
konfound	Omitted bias / cases	Regression inference
EValue	Unmeasured bias	Observational effects
boot	Sampling variation	Any statistic
influence.ME	Influential groups	Mixed models
car	Single-case influence	lm / glm
zaminfluence	Approximate deletion	Z-estimators

2 Related R packages

Every package in Table 1 answers a version of "how much would have to change for this conclusion to flip?", but they fall into two families.

sensemakr (Cinelli and Hazlett, 2020; Cinelli et al., 2024), **konfound** (Rosenberg, 2025) and **EValue** (VanderWeele and Ding, 2017) ask the question about *unmeasured* threats — a confounder, a selection mechanism, or unobserved cases the analyst never had. Their answers are analytic bounds computed from the reported estimate and its uncertainty; none of them touches the rows of the analysis dataset. **boot** (Canty and Ripley, 2025) is a general resampling engine rather than a conclusion-level robustness framework. **fragility** (Lin and Chu, 2025), **influence.ME** (Nieuwenhuis et al., 2012), **car** (Fox and Weisberg, 2019), `zaminfluence` (GitHub only; Giordano et al. (2026a), with the associated paper Broderick et al. (2023)) and **stabilitest** instead ask about the *observed* sample: which specific patients, if perturbed or removed, would change the conclusion?

Within that second family the differences are of scope and rigour rather than kind. **fragility** is restricted to flipping binary outcome labels, implementing the clinical fragility index of Walsh et al. (2014). **car** and **influence.ME** offer classical single-deletion diagnostics in the tradition of Cook (1977), which answer a different, single-deletion question from the package's multi-observation adversarial search. `zaminfluence` approximates the adversarial leave-k-out answer efficiently via influence functions, but for Z-estimators in general rather than the two-sample and model-term designs of trial reporting.

stabilitest's distinguishing choices relative to all of these are: it evaluates deletion, jackknife and bootstrap sensitivity together as three views of one conclusion rather than offering a single diagnostic; its worst-case removal is an actual adversarial refit search rather than a residual/leverage heuristic or a linear approximation; and its composite score carries a categorical verdict only where an independent pre-registered calibration has validated it for that exact configuration. No other package in Table 1 ships a calibration registry or a fail-closed labelling policy.

3 Methods

Given two samples — or a model term, see the API section — and a significance level α, four analyses are performed. For two independent groups, n_1 and n_2 denote per-group sizes and $N = n_1 + n_2$ is the total number of observations. Three analyses enter the composite score; the fourth is retained as a deliberate contrast.

Jackknife (leave-one-out). Each observation is deleted in turn and the test repeated, following the classical framework of Shao and Tu (1995). Outputs are the conclusion stability S_{jack} (percentage of deletions preserving the significance decision), the set of influential observations (deletion flips that decision, or shifts p by more than $\delta = 0.05$), and the leave-one-out p-value range. Because single-deletion influence shrinks as $\sim 1/n$, high jackknife stability in a large sample is *weak* evidence of robustness. The component's chief value is identifying specific observations for data-quality review; the package's printed output says so explicitly. Conclusion stability does not guarantee preservation of effect direction: direction reversals must be inspected separately from the significance decision.

Worst-case removal (primary fragility analysis). A greedy algorithm deletes, at each step, the

observation whose removal moves the p-value furthest toward overturning the conclusion, up to $k_{\max} = \lfloor 0.30\,n \rfloor$ deletions. The *removal fragility index* k_{frag} is the step at which the conclusion first flips. If no flip occurs by $k_{\max}$, the stored value $k_{\max}+1$ is a right-censoring code and is reported as $> k_{\max}$, not as an observed flip. This follows the maximum influence perturbation of Broderick et al. (2023) (see also Giordano et al. (2026b) and Giordano et al. (2026c)), who showed that in prominent published analyses removing under 1% of observations can reverse the sign of an estimated effect. When a flip is found, the greedy set is *sufficient* to overturn the conclusion, so k_{frag} is an upper bound on the size of the minimal overturning subset; no universal optimality claim is made. Because a flip from $p = 0.047$ to $p = 0.053$ is a knife-edge event (Potter, 2020), the package returns and plots the full p-value trajectory rather than the index alone. Note the terminology: the *removal* fragility index counts deleted observations and so reduces n, which distinguishes it from the Walsh fragility index (Walsh et al., 2014), which flips event status while holding n fixed.

Extreme-value removal (descriptive). Cumulative deletion of observations most distant from the recomputed grand mean — the familiar "we removed the outliers and the result held" procedure. It does *not* enter the composite score. It is retained because the gap between this index and the worst-case index measures how much protection the outlier narrative actually provides.

Bootstrap same-decision rate. B resamples are drawn with replacement within groups (Efron and Tibshirani, 1993); S_{boot} is the empirical proportion that preserves the observed significance decision. It is a plug-in quantity under the empirical distribution, not an unconditional probability that a future study will replicate. It mostly reflects strength of evidence at the observed effect size, not resistance to contamination: a marginal p-value can yield S_{boot} near 50–60% in clean data. The API retains the historical field name `bootstrap_reproducibility` for compatibility.

Composite score. $R = w_1 S_{\text{jack}} + w_2 F + w_3 S_{\text{boot}}$, where $F = 100 \cdot \min\left(k_{\text{frag}}/(k_{\max}+1),\, 1\right)$ rescales the fragility index to the full 0–100 range. Default weights are $w = (0.4, 0.4, 0.2)$: influence and fragility carry equal weight, the same-decision term is down-weighted because it largely restates the p-value. The weights are a heuristic communication device, not a derived quantity. They are an explicit argument and should be fixed in the statistical analysis plan. Two results with the same score can be stable in different ways, so the components — especially the fragility index and the identity of the observations behind it — are the substance of reporting.

4 Package design and API

Dispatcher and engines

Two-sample analyses go through a single dispatcher; model-based and equivalence analyses have their own entry points sharing a common case-deletion engine.

```
# Two-sample: t.test, paired.t.test, wilcoxon, wilcoxon.paired,
#             brunner_munzel, fisher, chisq, prop
robustness_analysis(group1, group2, test_type = "t.test")

# Model terms: single coefficient or a multi-df factor tested jointly
robustness_lm(change ~ arm + baseline, data = d, term = "arm")
robustness_glm(y ~ arm + age, data = d, term = "arm", family = binomial())
robustness_surv(Surv(time, event) ~ arm, data = d, term = "arm")

# Equivalence / non-inferiority: mean, risk difference, odds ratio
robustness_tost(g1, g2, type = "equivalence", endpoint = "mean",
                delta_L = -20, delta_U = 20)
```

For model engines all components operate on *rows* (subjects) of the analysis dataset, with the conclusion defined by the p-value of a pre-specified term — a single coefficient, or a multi-df factor term via a joint test (`drop1` F for `lm`, likelihood-ratio for `glm` and `coxph`). Deleting whole subjects, with their covariates and follow-up, preserves the adjustment structure and handles censoring naturally. Fits that fail after deletion (a factor level vanishing, say) are skipped and reported rather than silently dropped.

Arguments are deliberately uniform across engines: `alpha`, `n_boot`, `max_removal_pct`, `weights`, and `seed` mean the same thing everywhere, and `seed` defaults to `123` so that a bare call is reproducible.

Return objects and methods

Analyses return a classed list (`robustness_analysis`, `robustness_model`, or `robustness_tost`) containing the original test result, a `robustness_metrics` tibble of 17 component metrics, the per-step trajectories of each removal analysis, the bootstrap draws, the resolved calibration decision, and the analysis profile used to make that decision. `print()` methods render a structured summary; `plot()` returns the removal trajectories and the bootstrap p-value distribution.

The calibration registry

The mechanism that makes label suppression systematic rather than ad hoc is a registry shipped as `inst/extdata/calibration-registry.csv`, keyed by the *resolved* method rather than by the user-facing argument. A call with `test_type = "t.test"` resolves to `welch_unpaired` or `paired_t`; one with `test_type = "fisher"` resolves to `fisher_exact`. Each row carries a status, cutoffs where applicable, a version string, and a provenance pointer to the study that produced it.

Label resolution is fail-closed. A categorical label is emitted only when the resolved method has a `validated_method_specific` row **and** the observed conclusion type matches **and** the weight vector matches the calibrated one **and** the design profile falls inside the study's eligibility bounds. Any mismatch yields `NA` — with a printed reason — while the numeric score and all components are retained. There is no fallback to a "nearest" calibration. The historical Welch 55/70 mapping is now an inactive comparator and is never applied to Welch, ANCOVA, or another runtime result.

5 Calibration policy and evidence

The Welch unit

The original 55/70 mapping was based on differences between scenario-level average scores. A prospective replacement study tested whether those averages supported case-level categorical interpretation. The calibration unit was a significant independent-groups Welch test at $\alpha = 0.05$, with default weights $(0.4, 0.4, 0.2)$, `n_boot = 1000`, and a 30% removal cap. The six core design archetypes crossed per-group sizes from 25 to 200, balanced and 40/60 allocations, and variance ratios 1, 2, and 1/2. Within each archetype the null used zero mean difference, the borderline stratum targeted 60% power, and the clear stratum targeted 95% power.

All 18 training cells reached their quota of 250 completed significant analyses: 1,500 null, 1,500 borderline, and 1,500 clear analyses (4,500 total), with zero analysis failures. The frozen search considered integer three-band and two-band mappings. No candidate passed all safety, identification, ordinal-accuracy, occupancy, and ordering requirements, so the final result is `uncalibrated` with no cutoffs. Under the pre-specified protocol this training outcome was final: held-out validation remained unopened, no validation estimates exist, and no positive runtime eligibility profile was published.

The historical 55/70 mapping was retained as a named training comparator. Its false-reassurance estimate was 0.458 (Wilson upper 0.479, scenario-cluster upper 0.513, conservative upper 0.513), against gates of 0.05 and 0.10. Clear-effect identification was 0.499 (Wilson lower 0.478, scenario-cluster lower 0.348, conservative lower 0.348), against gates of 0.70 and 0.60. Balanced ordinal accuracy was 0.558 (gate 0.70), and minimum band occupancy was 0.227. Pooled ordering was preserved and no material reversal was detected, but those secondary conditions could not rescue the failed primary gates.

The frozen no-candidate SHA-256 is represented in eight consecutive blocks. The first four are `9c45481b` `952cab7c` `b9b9086e` `37924a39`; the final four are `d83fe048` `4628745d` `be2e79eb` `33e8797d`. the compact supplementary artifact records this hash, source hashes, runtime, occupancy, and the unopened-validation flag.

A relaxed-gates replication

A reader may reasonably ask whether the failure came from gates that were too strict. A second prospective study (`welch-2026-3`) therefore repeated the design with fresh scenario identities and random streams and a deliberately weaker claim. It looked for one cutoff that earns a single `Stable` label, accepting false reassurance up to 0.10 (conservative upper bound 0.15) and clear-effect identification down to 0.60 (conservative lower bound 0.50). Any cutoff also had to satisfy those gates at its two neighbours on each side, false reassurance had to stay at or below 0.25 within every archetype, and the score had to beat a p-value-only rule fitted by the same procedure. The relaxed gates were chosen

Table 2: Descriptive historical Welch stress simulation, conditional on statistical significance; not training or confirmation evidence for the prospective calibration decision. There are 500 replications per scenario and B = 200. Columns report effect size, per-group n, injected contaminating observations, rejection rate, mean score among significant results, median worst-case and extreme-value fragility indices, and mean bootstrap same-decision rate.

d	n	Cont.	Rej. rate	Score	k_{wc}	k_{ex}	Boot. %
0.0	25	0	0.040	54.7	2.0	2	67.1
0.0	25	2	0.064	52.6	1.0	1	66.6
0.0	50	0	0.076	50.0	1.5	2	63.1
0.0	50	2	0.084	51.3	2.0	2	65.5
0.5	25	0	0.392	61.1	3.0	4	75.3
0.5	25	2	0.670	64.5	4.0	4	80.2
0.5	50	0	0.698	63.2	5.0	11	80.3
0.5	50	2	0.864	65.7	6.0	10	84.8
0.8	25	0	0.808	69.7	5.0	8	83.0
0.8	25	2	0.922	73.3	6.0	9	88.6
0.8	50	0	0.976	75.5	12.0	26	94.1
0.8	50	2	1.000	77.6	14.0	26	96.3

after seeing the first study's training rows, which is why the new study used new data and froze its protocol before producing any score.

Training again completed all 18 cells (4,500 significant analyses, 0 failures). The score did find a cutoff: 65. At that cutoff false reassurance was 0.048 (conservative upper 0.093) and clear-effect identification was 0.703 (conservative lower 0.603), although false reassurance reached 0.164 in the smallest archetype (W01, 25 per group). The comparison with the p-value is what closed the study. A rule that labels results with $p \leq 0.002$ had false reassurance 0.035 and clear-effect identification 0.713 on the same rows, no worse than the score. The p-value rule also had no eligible cutoff under the neighbourhood requirement (only 3 of 10 grid values met the gates), and the frozen protocol treats that as a closed outcome. Because the unavailable comparator was known from training alone, held-out validation was not opened. The candidate hash is b113e704 766eb8d0 740af9c5 f387044b 175c0d0b 078d971f b867bd92 8c0b4f98. The active registry version is now `welch-2026-3`: numeric scores and components remain available, while the categorical label is `NA`. On these training rows the score offered no advantage over a p-value cutoff at comparable error rates, so a categorical label would have summarized little beyond the p-value.

Table 2 retains the earlier 12-scenario experiment as a descriptive stress example, regenerated with the released package (12 scenarios $\times$ 500 replications, $B = 200$). It was not used to fit or confirm the prospective decision.

Two features of Table 2 deserve emphasis. First, under the null the rejection rate is at or near nominal in clean data (0.040 and 0.076 across the two sample sizes) and is higher under contamination (0.064 and 0.084). The Monte Carlo standard error at 500 replications is about 0.010, so individual cells should not be over-read. Among findings that reached significance, median worst-case fragility is one or two observations throughout the null rows. This is a descriptive pattern, not evidence that a particular score threshold safely classifies an individual result.

Second, and more consequentially, the median extreme-value index is systematically larger than the worst-case index in the $d = 0.8$ scenarios — 26 versus 12 at $d = 0.8$, $n = 50$. Judging stability by removing apparent outliers makes a result look roughly twice as sturdy as an adversarial reading shows it to be. The most damaging observations are usually not the obvious ones.

Figure 1 shows the underlying score distributions.

The Fisher unit: a calibration earned

An independent study for Fisher's exact test on two-arm binary outcomes delivered the package's sole active categorical calibration. Its design was frozen before execution: truth strata defined by *enumerated exact power* rather than by observed effect size (null exact; clear at exact power 0.95; a borderline stratum at 0.60 retained as diagnostic only and excluded from threshold fitting), and a sealed score-only pilot gating production.

The study also settled the weight question empirically. On binary data the jackknife stability

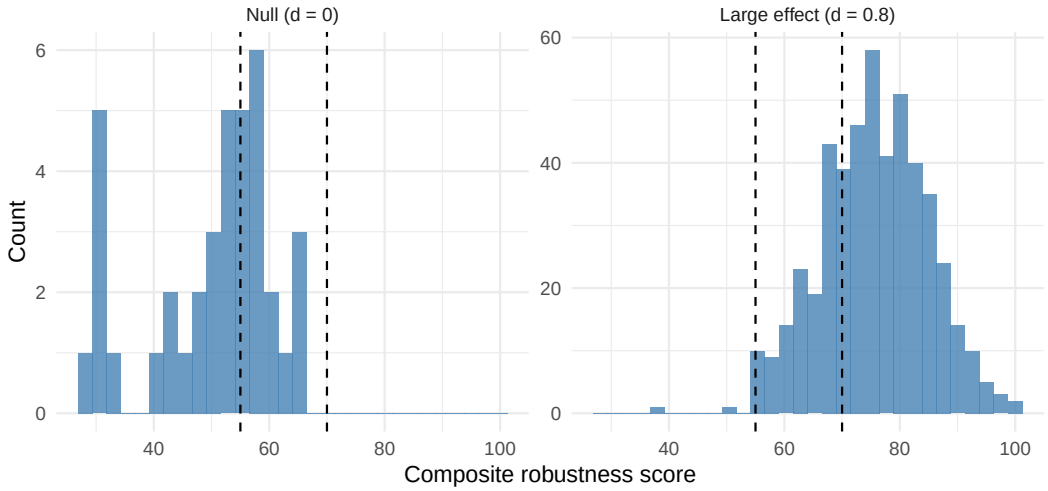


Figure 1: Descriptive score distributions for significant results under the null (d = 0, left) and the study's d = 0.8 scenario (right), n = 50 per group, uncontaminated. Dashed lines mark the inactive historical 55/70 mapping. The distributions overlap; this figure is not validating evidence for categorical bands.

component is constant at 100 in every cell — a single deletion cannot move a Fisher p-value across α at these sample sizes — so it carries zero information. The frozen design therefore uses *jackknife-light* weights $(0, 0.5, 0.5)$, which maximise score granularity without changing discrimination.

Training selected the cutoff $L = 58$ with false reassurance 0.0493 (Wilson upper 0.0607) and identification of clear effects 0.7608 (Wilson lower 0.740). Held-out confirmation, opened only once, gave 0.0367 and 0.7333; under the conservative-of-Wilson-and-cluster-bootstrap rule the reported bounds are a false-reassurance upper of 0.0617 and an identification lower of 0.6767, both inside the frozen gates. The registry row is `fisher-2026-1`, provenance `study:binary_proportion@cc3344931614`.

The Fisher study was calibrated before the Welch replication added a p-value-only baseline, so a descriptive comparison was run afterwards on the published rows. The comparison rule was fixed on training: the p-value cutoff on a coarse grid whose training false reassurance is closest to the score's, here $p \leq 0.002$. In training that rule had false reassurance 0.025 and clear-effect identification 0.728, against 0.049 and 0.761 for the score. The score identified 3.3 percentage points more clear effects, with a 90% scenario-cluster interval from -1.2 to 8.3, and at higher false reassurance. In the held-out rows, read here descriptively and not used for any selection, the two rules gave 0.037 / 0.733 (score) and 0.027 / 0.700 (p rule), and the areas under the curve were nearly equal (0.919 and 0.920). The validated error rates describe the label as published. They do not show that the score separates clear effects from null results better than a p-value cutoff does.

The resulting mapping is two-band — score ≤ 58 *Fragile*, > 58 *Not fragile* — with **no Robust tier**, because a single binary comparison does not carry enough information to support a third band at the required error rates. It applies only within the study's eligibility profile: per-arm n in $[25, 200]$, allocation ratio in $[0.8, 1.25]$, control-arm event rate in $[0.08, 0.55]$ with at least three events and three non-events. Outside that profile, or under default weights, the label is suppressed.

ANCOVA: a named negative result

Covariate-adjusted analysis is an important calibration target because many confirmatory trials adjust for baseline. Three pre-registered studies were conducted; all closed without a usable label under their frozen designs.

ANCOVA Study 1 evaluated a three-band mapping with default weights. No threshold met the pre-specified training requirements, so held-out data was not opened. Within its canonical Gaussian scenarios the composite score was nearly monotone in the p-value (Spearman correlation between its bootstrap component and $-\log_{10} p$ was approximately 0.99). The result establishes empirical infeasibility under that design; it does not prove that every future ANCOVA score or scenario definition must fail.

ANCOVA Study 2 evaluated a two-band mapping with jackknife-light weights. Its sealed pilot showed score-location separation ($\Delta = 24.4$, overlap 0.034, AUC 0.892), but production training found no feasible cutoff: the best clear-effect identification at a false-reassurance-safe threshold was 0.554, below the pre-specified 0.70 requirement. This discrepancy shows why pilot location metrics alone are

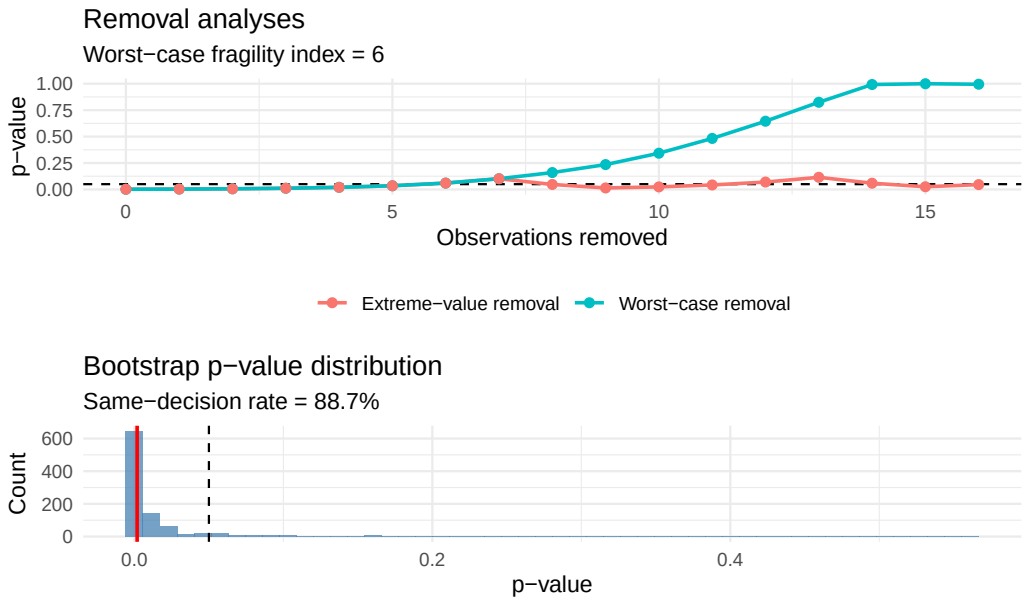


Figure 2: Diagnostic plots for the Welch case study. Top: p-value trajectories under worst-case (adversarial) and extreme-value (outlier-ranked) removal, with the dashed line at alpha = 0.05. Bottom: bootstrap p-value distribution, red line at the observed p-value.

insufficient for deciding whether a categorical claim is feasible.

The **assumption-violation study** then asked whether the score added discrimination beyond p for violations represented in a frozen scenario suite. It completed 2,100 analyses with zero failures. The pooled difference $\mathrm{AUC}_{\mathrm{score}} - \mathrm{AUC}_p$ was 0.0053, with a scenario-cluster bootstrap 95% interval $[-0.111, 0.129]$; the added discrimination was not confirmed.

The ANCOVA registry rows remain `uncalibrated`, and labels stay suppressed. Audit identifiers, hashes, and raw decision strings are retained in the study-specific `published/` directories rather than in reader-facing prose.

6 Examples

Welch case study

The package ships a synthetic phase II analgesic trial: 55 patients, change from baseline in pain score at week 12. Results below are loaded from a committed artifact generated with the package-default `n_boot = 1000` and `seed = 123`. Welch is numeric-only after the prospective negative result.

```
res <- robustness_analysis(pain_treatment, pain_placebo, test_type = "t.test",
                           n_boot = 1000, seed = 123, interpret = TRUE)
```

The Welch test gives a mean difference of -11.40 points, $p = 0.0018$. The robustness analysis returns a numeric score of **71.9/100** with no categorical label, built from: jackknife stability 100.0% with 0 influential observations flagged; a worst-case fragility index of $k = 6$ (10.9% of the sample), with p rising to 0.0602 after the sixth deletion; and bootstrap same-decision rate 88.7%.

Figure 2 shows the two diagnostic displays produced by `plot()`. The upper panel makes the contrast of the Methods section concrete: the worst-case trajectory crosses α well before the extreme-value trajectory does.

For a reader, the actionable output is not the score but the six named patients — led by one extreme responder — whose charts warrant source-data review. The observed component pattern resembles the study's clear-effect scenarios more than its null scenarios, but a sensitivity diagnostic does not establish which data-generating truth produced an individual dataset.

Other engines

Short live examples across the remaining engines. An ANCOVA term on the packaged illustration dataset:

```
res_lm <- robustness_lm(change ~ arm + baseline_pain, data = pain_ancova_trial,
                        term = "arm", n_boot = 300, seed = 123)
res_lm$robustness_metrics[, c("worstcase_fragility_k", "worstcase_fragility_pct",
                              "bootstrap_reproducibility", "overall_robustness")]
```

```
#> # A tibble: 1 x 4
#>   worstcase_fragility_k worstcase_fragility_pct bootstrap_reproducibility
#>                   <int>                   <dbl>                     <dbl>
#> 1                     7                    8.75                      83.3
#> # i 1 more variable: overall_robustness <dbl>
```

The score is numeric-only: 67.9 with no label, exactly as the negative result above requires. A TOST equivalence analysis behaves the same way:

```
res_tost <- robustness_tost(pain_treatment, pain_placebo, type = "equivalence",
                            endpoint = "mean", delta_L = -20, delta_U = 20,
                            n_boot = 300, seed = 123)
c(fragility_k = res_tost$robustness_metrics$worstcase_fragility_k,
  score = round(res_tost$robustness_metrics$overall_robustness, 1))
```

```
#> fragility_k       score
#>         4.0        65.4
```

A Fisher analysis inside the calibrated profile, under the calibrated weights, does receive a label:

```
set.seed(2026)
ctrl <- rbinom(60, 1, 0.15); trt <- rbinom(60, 1, 0.40)
res_f <- robustness_analysis(trt, ctrl, test_type = "fisher", n_boot = 1000,
                             seed = 123,
                             weights = c(jackknife = 0, fragility = 0.5,
                                         bootstrap = 0.5))
c(score = round(res_f$robustness_metrics$overall_robustness, 1),
  label = res_f$robustness_interpretation)
```

```
#>         score         label
#>        "67.6" "Not fragile"
```

The same data under default weights returns identical component metrics but a different composite (75.7) and a suppressed label — the registry requires the calibrated weight vector, not merely the calibrated method.

7 Performance

Cost is dominated by the greedy search, which performs $O(N \cdot k_{\max})$ test evaluations. For total $N = 55$, $k_{\max} = 16$ and the greedy candidate refits total $\sum_{j=40}^{55} j = 760$. A full default run uses approximately $1 + 55 + 760 + 16 + 1000 = 1832$ test evaluations before overhead: the original test, jackknife, greedy search, descriptive extreme-value path, and bootstrap. Closed-form tests are therefore sub-second; model engines pay for repeated refits.

Table 3: Wall-clock seconds for a single analysis with 1000 bootstrap resamples, stabilitest 0.6.5 on an Apple aarch64 laptop. The Cox benchmark uses simulated survival data generated in the timing script.

Engine	n = 50 per arm (s)	n = 150 per arm (s)
robustness_analysis (Welch)	0.44	1.25

robustness_lm (ANCOVA)	0.89	3.63
robustness_glm (binomial)	1.36	6.74
robustness_surv (Cox)	1.32	4.85
robustness_tost (mean equivalence)	0.67	1.28

The practical implication is that interactive use is comfortable at trial scale, and that a large calibration study — thousands of replications — belongs in a batch script, which is how the studies reported here were run.

8 Testing and quality assurance

The package carries 1090 passing `testthat` assertions across 9 test files, covering the analysis engines, edge cases (degenerate groups, zero variance, complete separation, failed refits), the packaged datasets, the calibration registry's resolution logic, and the calibration documentation itself.

That last one is unusual enough to describe. A source-tree documentation audit, run as part of the test suite, asserts that the README, NEWS, vignettes, manuscript, and every study SAP still state the policy invariants: that uncalibrated labels are suppressed, that the historical Welch mapping is inactive, and that each frozen acceptance criterion remains present in its own statistical analysis plan. The intent is to make it mechanically hard to quietly weaken a published calibration claim through documentation drift.

Each calibration study additionally publishes compact decision artifacts in its own `published/` directory: occupancy and failure tables, power verification, candidate diagnostics, a registry row, manifests, and a hash ledger. Candidate hashes are recorded before held-out data is opened, so the sequencing of any decision is auditable after the fact.

9 Discussion and limitations

The framework separates three questions usually blurred together, and its sharpest practical lesson is the gap between adversarial and symmetric removal: in the descriptive stress simulation, adversarial deletion and cumulative grand-mean extreme-value deletion produced materially different fragility summaries. That comparison does not benchmark Cook's distance, DFBETAs, **car**, or **influence.ME**.

Several limitations deserve statement rather than burial.

The score is largely a monotone transform of the p-value. In clean parametric settings the composite score re-expresses the evidence the p-value already carries; the ANCOVA program measured this directly. This is a scope condition, not a defect, but it bounds what the score can be used for. Its added value lies in (i) identifying the specific observations carrying the conclusion, (ii) the worst-case deletion bound, and (iii) behaviour under contamination, where the p-value alone is silent — see the null-with- contamination rows of Table 2. The assumption-violation study tested whether the score added discrimination beyond p under its frozen violation suite and did not confirm such an improvement.

The weights are a convention. They are explicit, pre-specifiable, and demonstrably not load-bearing for discrimination (the Fisher weight study found AUC essentially unchanged across weightings), but they are not derived from theory. The score is a communication device; the components are the substance.

The greedy index is an upper bound. Exact minimal overturning subsets are combinatorially hard. The algorithm returns a sufficient overturning set when it finds one, not a certificate of minimality. Certified minimal sets — via integer-programming formulations — remain future work.

Binary significance framing. Inherited from the fragility concept itself and criticised for good reason (Potter, 2020); the package's response is to report the trajectory alongside the index and to frame the index as an upper bound, wherever the index is defined rather than only in a limitations section.

Independence is assumed. Clustered and longitudinal designs are not supported; for those, `influence.ME` addresses the grouping-factor question.

Calibration coverage is deliberately narrow. Only the eligible Fisher exact-test configuration under jackknife-light weights carries a label. Welch, paired, rank, Brunner–Munzel, non-canonical binary, model-based, Cox, and TOST analyses return numeric output with a suppressed label pending an independent positive calibration. Users who want a verdict where none is licensed are, by design, not given one.

10 Summary

stabilitest asks how much of the data a conclusion actually rests on, and answers with component metrics that name the observations responsible. Its methodological position is that a categorical robustness verdict is a claim requiring validation for the specific analysis at hand, and that the honest response to a failed validation is silence plus a published negative result. The package is available from github.com/ma-brain/stabilitest.

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